

DMG / CYD Game Boy - Wiring Reference

ESP32-2432S024 | 2.4in 240x320 | board shown from the BACK | I2C entirely on CN1 | onboard amp, no external amp | rev C

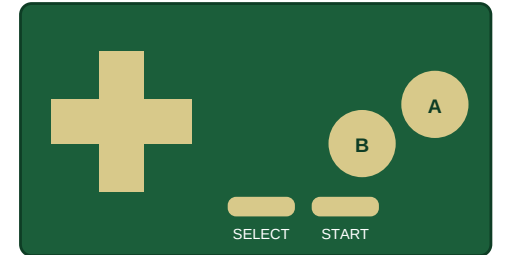
PN532 NFC READER (I2C mode)

Set the DIP switches to I2C. Address 0x24.
VCC 3.3V GND SDA SCL - all from CN1.
Read once at boot / cart insert only.
NTAG215 UID + NDEF text record -> ROM filename.
Remove the DMG RF shield or the read will fail.
The DMG interlock blocks cart removal with power on,
so every cart change is followed by a boot anyway.

MCP23017 I2C GPIO EXPANDER

A0/A1/A2 -> GND (address 0x20)
RESET -> 3.3V (never leave it floating)
 VDD 3.3V VSS GND SDA SCL - all from CN1.
 GPPU = 0xFF (internal pull-ups), buttons active LOW.
 INT unused - poll once per frame, ~60 us at 400 kHz.
 Port B is entirely free.

BUTTON PCB (SNES layout)



Bottom 8 header pins run in order to A7 -> A0.

UP -> A7 START -> A3
DOWN -> A6 SELECT -> A2
LEFT -> A5 A -> A1
RIGHT -> A4 B -> A0

GND bridged separately from the PCB header to the module GND.

X, Y, L, R pads are unused - a DMG shell has no buttons for them.

Debounce ~8 ms in software; the expander has no hardware filter.

Combos:

- Start+Select menu
- Select+Up/Down volume
- Select+Left/Right brightness

POWER

LiPo (+) -> SWITCH -> BAT +
LiPo (-) -----> BAT -

3.7 V pouch, ~2000 mAh, WITH a protection PCB. Approx 60x35x7 mm.

BRIDGE SW1's TWO PADS.
SW1 is a momentary that latches the battery rail on. It has no off path, so without a bridge the unit will not start inside a sealed shell.

CHARGE WITH THE SWITCH ON.
The switch cuts the cell from the whole board, onboard charger included.

USB-C is reached through a slot in the shell; the board sits 4 mm toward the USB side (GAME_X compensates).

Meter polarity before first connect.

SPEAKER 2-pin JST

8 ohm, ~36 mm. Reuse the original DMG driver.

Onboard amp tested OK - no external amp needed.

IO4 is NOT an amp enable.
The vendor datasheet is wrong; mute via IO4 was tested and does nothing.

No hardware mute exists, so volume is software only. For OFF, hold the DAC at 128 (mid-scale, not 0).

WHAT CHANGED (rev C - all bench-verified)

1. **I2C is entirely on CN1. GND / IO22 / IO27 / 3.3V on one 4-pin 1.25 mm connector. The earlier IO27+IO1 arrangement is gone,** and with it the bootloader-noise reasoning, the nine-pulse bus-recovery routine and the never-use-IO3 warning. Nothing is soldered to the CYD.
2. **IO4 is not an amp enable. Tested: the mute command printed but audio continued. The vendor datasheet is wrong about this pin.** No hardware mute, so auto-mute-on-silence is not possible and volume OFF must hold the DAC at mid-scale.
3. **Onboard audio is good enough. Verified with real Game Boy music from SD, on battery. No resistor mod, no MAX98357A, three GPIO stay free.**
4. **SW1 is a latching power button, not a switch. One press after the battery is connected; pressing again does nothing. Bridge its pads.**
5. **P3 (GND / IO35 / IO22 / IO21) is spare. IO21 is the backlight - leave it alone. IO35 is input-only.**

The vendor datasheet has now been wrong about the header pinout and about IO4. Meter anything from it before building on it.

WIRE COLOUR KEY

-  SDA IO22 (CN1)
-  SCL IO27 (CN1)
-  3.3 V (CN1)
-  GND (CN1)
-  Battery + via switch
-  Speaker +/-
-  Buttons A7-A0
-  Solder bridge

STILL TO VERIFY

Brownout voltage - sweep the cell 4.2 -> 3.3 V, note where the display quits.

IO34 divider ratio, against a meter.

SW1 bridge - hold it while connecting the battery. If it boots, a permanent bridge is safe.

